

CALM Stylized Facts

Kraken BTC/USD L3 Evidence for TCA Simulation

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Scope and Priority Labels

Goal. Calibrate and validate a limit-order-book simulator whose output is a *liquidation impact* estimate. Priority follows how directly a fact changes that estimate.

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- | | |
|-----------|---|
| P0 | You configure it, or it is the answer. Either a parameter typed into the simulator, or the number the simulator must reproduce. Seven facts. Wrong here and the liquidation estimate is wrong by construction. |
| P1 | It must emerge correctly. Not set directly, but if the simulated market fails to reproduce it the cost number cannot be trusted. Six facts. |
| P2 | Realism and second order. Improves face validity and matters for multi-liquidation cascade scenarios; little effect on a single liquidation cost. Six facts. |
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Calibration budget. Nineteen facts is more than one project can calibrate. If effort is constrained, the ten-fact core is **all of P0 plus facts 8, 9 and 10** — the flow-to-price transfer, the per-order impact kernel, and order-sign memory.

Why this ordering. Taken from the capstone proposal: the deliverable is a discrete-event LOB simulator with market-maker, taker and liquidation agents, used to quantify a market impact coefficient for TWAP / VWAP / instant liquidation under different liquidity regimes, and to answer two named questions — *liquidity capacity thresholds* and *spread recovery times*. Facts that feed those two answers, or that set a named agent parameter, are P0.

Measurability. Facts 1–3 have no parent-order labels in public data and are measured through a *proxy*; they are validation targets, not calibration inputs.

Proposal Requirements Mapped to Facts

Proposal component	Facts that calibrate or validate it
Taker bots, specified as <i>Poisson arrival times</i>	19 arrivals (Poisson is rejected : Fano 3 at 1 s rising to 28 at 300 s), 16 order sizes, 10 sign memory
Market makers, Avellaneda–Stoikov with inventory	6 spread, 13 queue imbalance, 20 depth profile, 7 resiliency
Liquidation bot: TWAP vs VWAP vs instant	1–3 impact and its duration exponent γ , 7 recovery between slices, 22 the market's own execution clock
Market impact coefficient under liquidity regimes	8 OFI, 9 response function, 5 cost curve; normal against stress in the appendix
Named question: <i>liquidity capacity thresholds</i>	5 cost curve and visible depth
Named question: <i>spread recovery times</i>	7 resiliency
Configurable parameters named in the proposal: <i>order arrival rates, book depth profiles</i>	19 and 20, both P0 for exactly this reason
Cascading liquidations	7 resiliency, 14 volatility clustering, 11 diffusivity

Data, Reconstruction and Proxy

Period. Normal week: 18–24 July 2026 (UTC), 167.95 measured hours. One extra calendar day precedes it purely to warm the book up.

Feed. Kraken BTC/USD **L3**: per-order add / modify / delete plus trade prints, top 100 price levels per side, tick = 0.1 USD.

- ▶ 85.6M raw rows $\rightarrow$ 61.7M packets $\rightarrow$ 161,702 aggressive orders, 4.49M top-of-book events, 396,688 fills.
- ▶ The window is depth-limited and scrolls out silently, so 23.4M level evictions are applied by the reconstruction; book state is read only at packet boundaries.
- ▶ Book and trade channels are separate subscriptions, so an order's pre-trade book is taken on *exchange* time. Aggressor-side consistency: 0.992 exchange-aligned against 0.777 receipt-aligned.
- ▶ 65 exchange snapshot blocks serve as out-of-sample checks; median reconstruction error 0 ticks.

Metaorder proxy (Facts 1–3). Public data carries no parent labels. Each aggressive order is assigned at random to one of $N = 6$ synthetic traders; a metaorder is a run of same-signed trades from one trader within a UTC day. Scales are per day: $V_D =$ daily volume, $\sigma_D = (\max p - \min p)/p_0$. Method: [Maitrier, Loeper & Bouchaud \(2025a\)](#), Algorithms 1–2. Three seeds are run; only $N = 6$ is reported here.

Convention. Signed impacts are kept throughout. No positive-impact pre-filter is applied, because selecting on the outcome changes the estimand.

P0 — Configure It, or It Is the Answer (1/2)

#	Fact	Definition	Why prioritised	Extant finding and source
1	Concave metaorder impact	$I(Q)/\sigma_D$ against Q/V_D at completion.	It is the deliverable. The proposal's "market impact coefficient" is this curve.	Square-root law on Bitcoin over six decades. Donier & Bonart (2015) .
5	Liquidity cost and depth	Virtual market-order VWAP cost and cumulative visible depth.	Named deliverable: the <i>liquidity capacity threshold</i> . Also bounds what one order can absorb.	Cost rises with size; book roughly symmetric. Schnaubelt et al. (2019) .
7	Liquidity resiliency	Recovery of spread, depth and cost after a large aggressive order.	Named deliverable: the <i>spread recovery time</i> . Sets safe spacing between slices.	Spread recovers over a few seconds. Schnaubelt et al. (2019) ; Obizhaeva & Wang (2013).
19	Event arrivals and clustering	Interarrival distribution and count dispersion.	Named configurable parameter (arrival rates), and the proposal's Poisson assumption is rejected by the data.	Strongly over-dispersed; self-exciting models are the standard fit. Bacry, Mastromatteo & Muzy (2015).

Facts 1–3: Metaorder Impact, $N = 6$ Proxy

Facts 1-4 proxy battery - random_n6 (seed 7) | Normal (18-24 Jul 2026)
Signed impacts retained throughout: no $I_{\text{norm}} > 0$ pre-filter

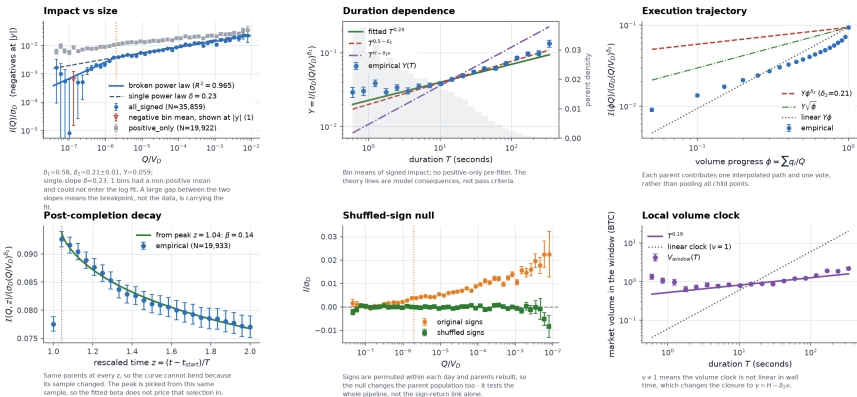


Figure 1: Same figure also carries facts 2 and 3, quantified two slides on. Impact is concave in size but far below square root: $\delta_2 = 0.21$ (single-slope $\delta = 0.23$), $Y = 0.061$. Decay exponent $\beta = 0.14$. The execution trajectory is linear rather than concave, $y(0.5)/y(1) = 0.39$ against 0.71 for $\sqrt{\phi}$.

Fact 5: Liquidity Cost and Visible Depth

Fact 5 - liquidity cost and depth | Normal (18-24 Jul 2026)

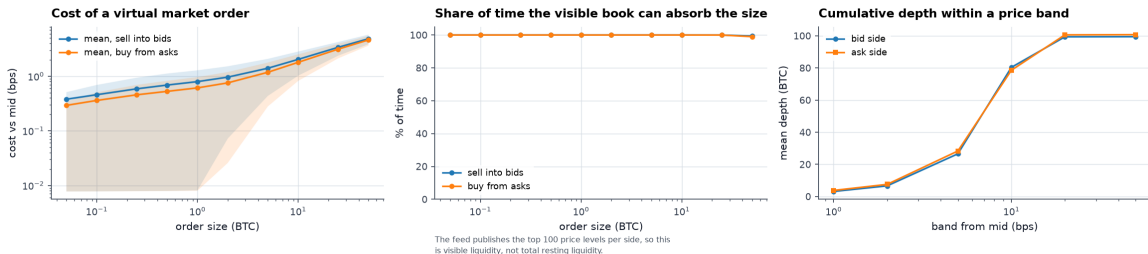


Figure 2: Virtual market-order cost rises steeply with size; cumulative visible depth saturates around 20 bps, which is where the 100-level publication window ends.

Fact 7: Liquidity Resiliency

Fact 7 - resiliency after large trades | Normal (18-24 Jul 2026)

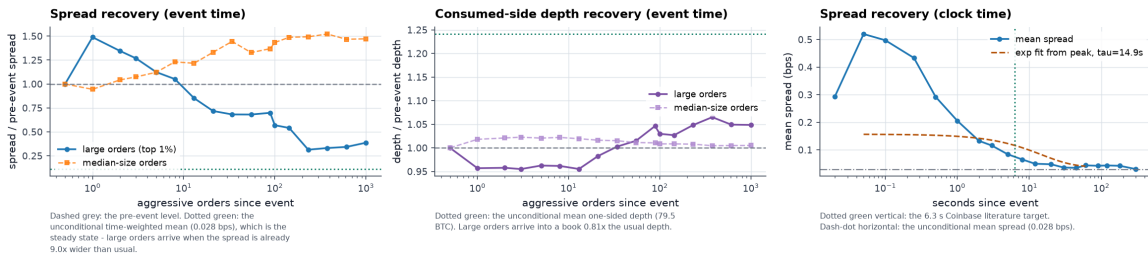
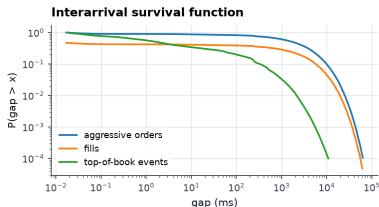


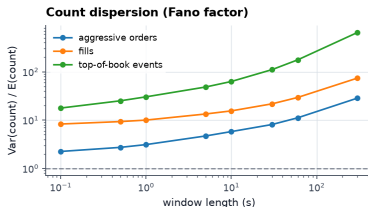
Figure 3: After large aggressive orders, spread and depth recover on different timescales.

Fact 19: Event Arrivals and Clustering

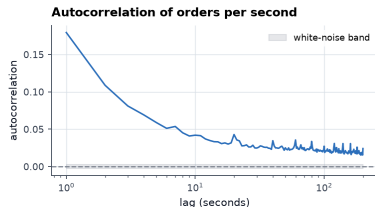
Fact 19 - event arrivals | Normal (18-24 Jul 2026)



A curve that starts below 1 has exactly-zero gaps: one aggressive order prints several fills inside the same nanosecond. The long flat plateau before the cliff is not a straight line, so these gaps are not a simple power law either.



1 is Poisson. Rising with the window is activity clustering - which a non-stationary intensity also produces, so this is description, not a Hawkes fit.



Slow decay is the activity-clustering counterpart of volatility clustering (fact 14).

Figure 4: Mean interarrival is 3.74s for aggressive orders and 135 ms for top-of-book events. The Fano factor rises from 3 at 1 s to 28 at 300 s: strongly over-dispersed against Poisson.

P0 — Configure It, or It Is the Answer (2/2)

#	Fact	Definition	Why prioritised	Extant finding and source
20	Book shape	L1 depth distribution and depth per level away from the touch.	Named configurable parameter (depth profiles). Sets how far a liquidation walks the book.	Depth grows away from the touch; sides roughly symmetric. Schnaubelt et al. (2019) .
16	Trade-size distribution	Distribution of aggressive-order and fill sizes.	Agent parameter: taker-bot order sizes. The tail governs how often the queue clears in one hit.	Heavy upper tail plus clustering on round sizes. Schnaubelt et al. (2019) .
6	Spread distribution	Distribution of the quoted spread and its state dependence.	Agent parameter: the maker's quoting width, and the unavoidable cost floor paid on every child order.	Tight with a heavy right tail; state dependent. Schnaubelt et al. (2019) .

Fact 20: Book Shape

Fact 20 - book shape | Normal (18-24 Jul 2026)

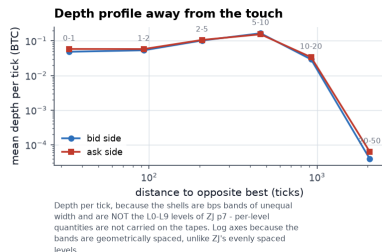
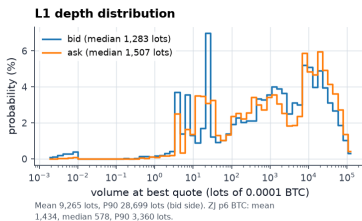
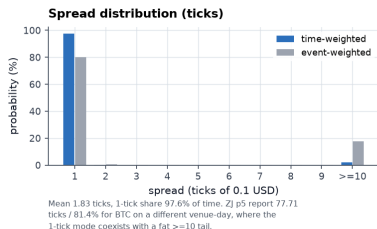


Figure 5: The book is one tick wide 97.6% of the *time* but only 80.2% of *updates*: wide spreads are short-lived (16 ms mean dwell against 164 ms at one tick).

Fact 20: L1 Depth and Volume Profile

Fact 20 - ZJ-style depth views | Normal (18-24 Jul 2026)

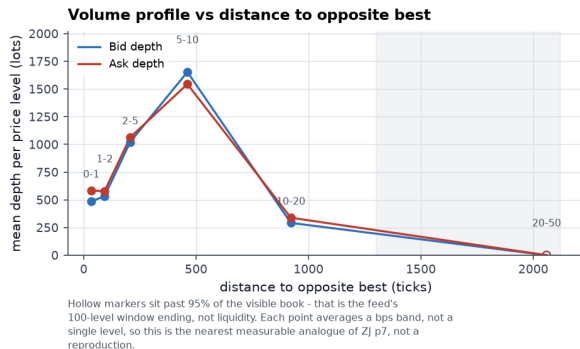
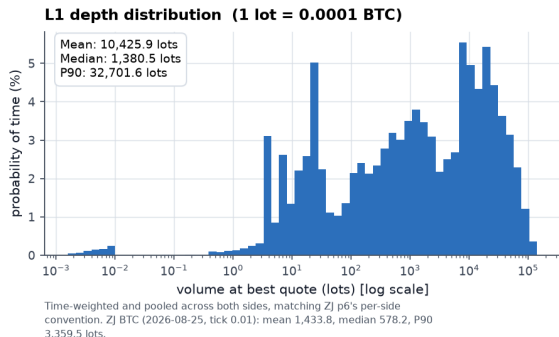


Figure 6: L1 depth median 1,380 lots (1 lot = 10^{-4} BTC). Depth per price level rises away from the touch and then collapses; hollow markers are the 100-level publication window ending, not the market thinning.

Fact 16: Trade-Size Distribution

Fact 16 - trade-size distribution | Normal (18-24 Jul 2026)

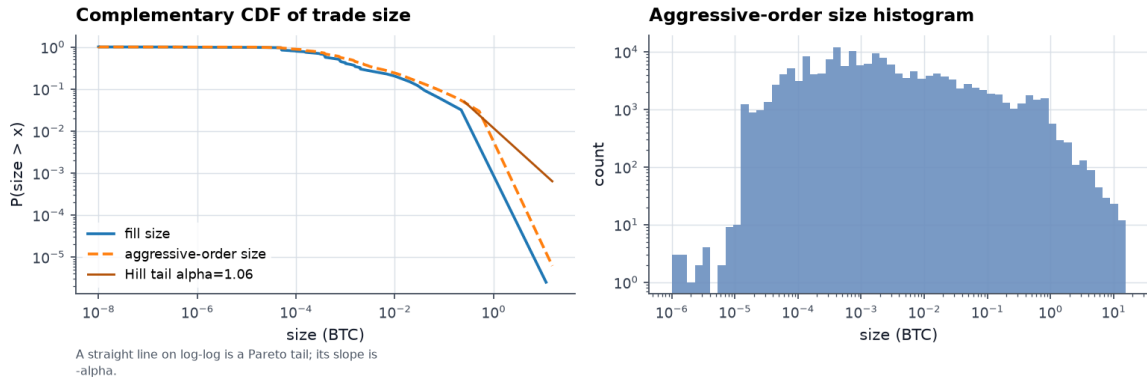


Figure 7: Aggressive-order sizes have a heavy upper tail: $p99/\text{median} \approx 700\times$ and the largest 1% of orders carry about 40% of all volume.

Fact 6: Spread Distribution and State Dependence

Fact 6 - spread distribution and state dependence | Normal (18-24 Jul 2026)

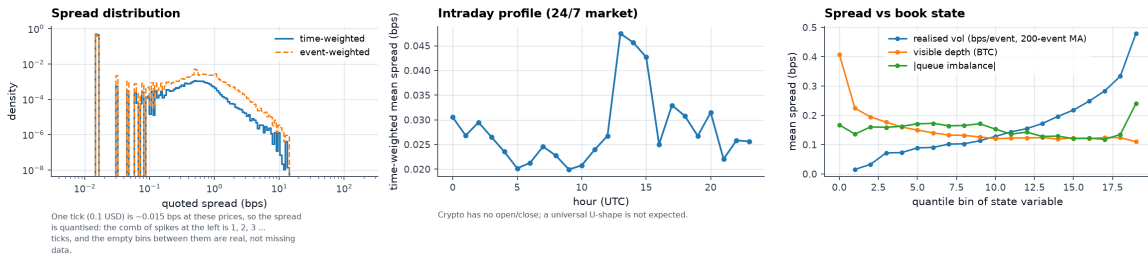


Figure 8: Spreads are tight and quantised: one tick is ≈ 0.015 bps at these prices, so the comb of spikes is 1, 2, 3 ticks and the gaps between are real.

P1 — Must Emerge Correctly (1/2)

#	Fact	Definition	Why prioritised	Extant finding and source
8	Order-flow imbalance	Mid-price change regressed on OFI over fixed windows.	The flow-to-price transfer function. If the simulated book does not reproduce it, the shock propagates at the wrong rate.	Mid changes approximately linear in OFI, coefficient falling with depth. Cont, Kukanov & Stoikov (2014) .
9	Trade response $\mathcal{R}(\ell)$	$\langle \epsilon_t(m_{t+\ell} - m_t) \rangle$ in trade time.	The per-order impact kernel. Metaorder cost is this aggregated over a schedule, so it must come out right first.	Rises with ℓ and settles despite persistent flow. Bouchaud et al. (2004) .
10	Order-sign long memory	$C(\ell) = \text{Cov}(\epsilon_t, \epsilon_{t+\ell}) \sim \ell^{-\gamma}$.	If simulated flow comes out i.i.d. the book refills too easily and liquidation cost is <i>understated</i> .	Power-law decay with $\gamma < 1$, from order splitting. Lillo & Farmer (2004) .

Fact 8: Order-Flow Imbalance, Event Time

Fact 8 - order-flow imbalance | Normal (18-24 Jul 2026)

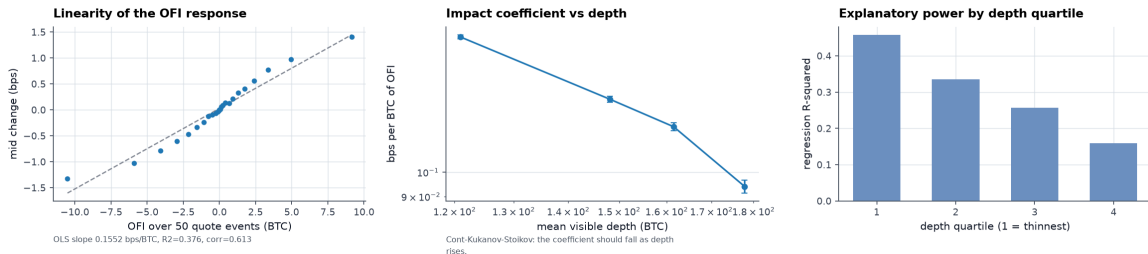


Figure 9: Mid change is close to linear in OFI over 50-quote-event blocks, and the impact coefficient falls as depth rises, as Cont, Kukanov & Stoikov predict.

Fact 8b: OFI versus TFI, Clock Windows

Fact 8b - OFI vs TFI | Normal (18-24 Jul 2026)

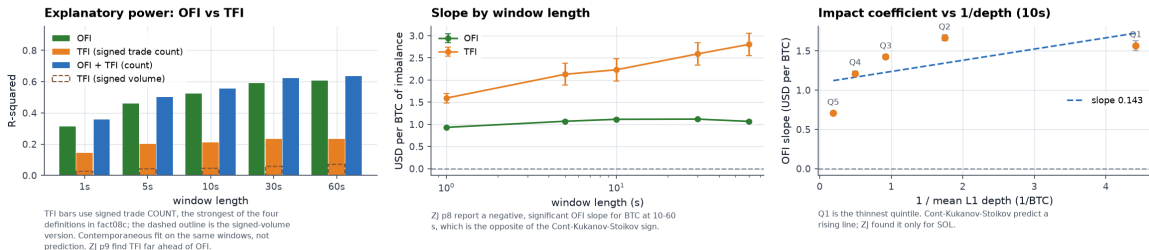


Figure 10: OFI explains $R^2 = 0.52$ at 10 s against 0.21 for the strongest trade-flow definition (signed trade count) and 0.05 for signed volume. The slope is positive and falls monotonically across depth quintiles.

Fact 8c: Real-Time OFI across Horizons

Fact 8c - real-time OFI and TFI definitions | Normal (18-24 Jul 2026)

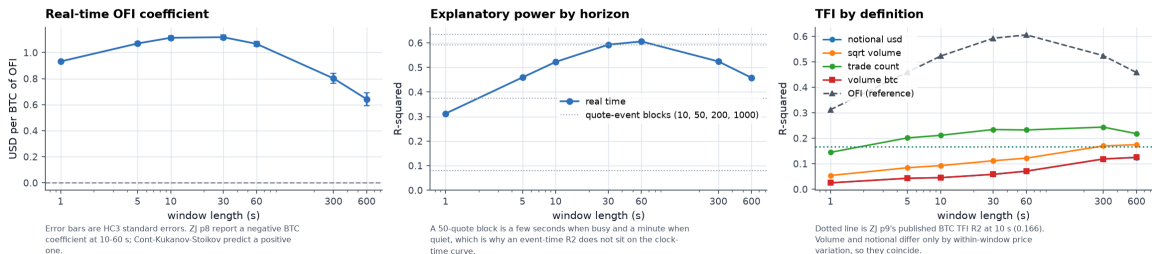


Figure 11: Explanatory power peaks near 60 s ($R^2 = 0.61$) and decays by 600 s (0.46); the slope peaks at +1.11 USD per BTC at 10–30s. The relation is a roughly one-minute phenomenon.

Fact 9: Trade Response Function

Fact 9 - trade response function | Normal (18-24 Jul 2026)

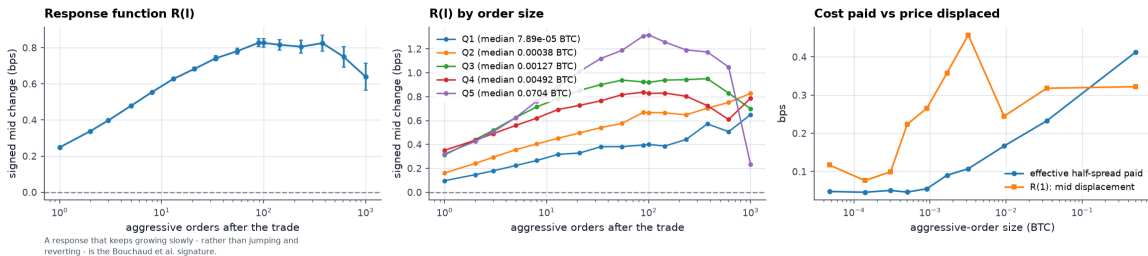


Figure 12: Signed aggressive orders move the mid in their own direction, growing slowly with lag rather than jumping and reverting.

Fact 9: Conditional Response $\mathcal{R}(\ell)$, $\mathcal{R}(v, 1)$, $\mathcal{R}^{\pm}(1)$

Fact 9b - conditioned trade response | Normal (18-24 Jul 2026)

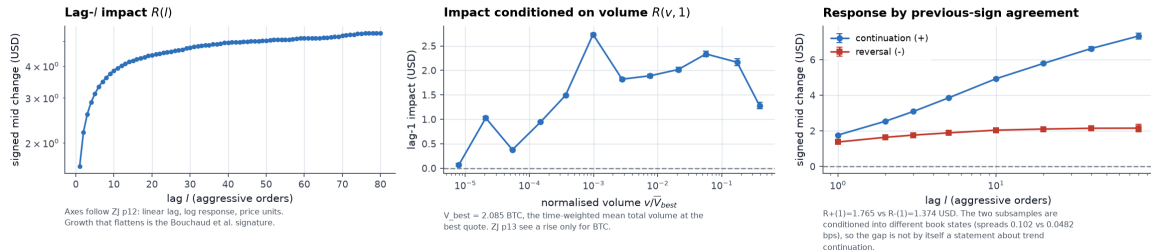


Figure 13: $\mathcal{R}(\ell)$ rises from 1.62 to 5.39 USD over 80 orders. $\mathcal{R}^+(1) = 1.77$ against $\mathcal{R}^-(1) = 1.37$ USD, but the two subsamples face different spreads (0.102 against 0.048 bps); scaled by the spread faced, the ordering reverses.

Fact 10: Order-Sign Long Memory

Fact 10 - long memory of order signs | Normal (18-24 Jul 2026)

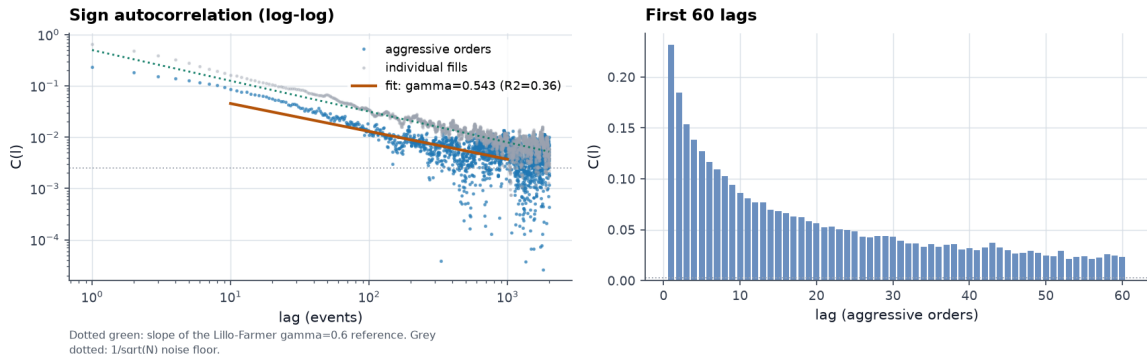


Figure 14: Signs decay as a power law with $\gamma = 0.54$ over lags 10–1000; $C(1) = 0.23$ and $C(100) = 0.011$, still above the noise floor.

P1 — Must Emerge Correctly (2/2)

#	Fact	Definition	Why prioritised	Extant finding and source
2	Impact trajectory and duration	Impact against executed fraction ϕ and execution time T .	Decides whether slowing down helps. TWAP beats an instant order only if impact depends on T .	Concave $\propto \sqrt{\phi}$; impact roughly independent of T . Maitrier et al. (2025a) .
3	Post-execution decay	Relaxation of impact after the last child order.	Splits cost into temporary and permanent. Only the permanent part propagates into a cascade.	Decays toward a permanent level; propagator picture. Bouchaud et al. (2004) .
11	Diffusive prices under persistent flow	Variance ratio and return autocorrelation across clocks.	No-free-money check. Persistent flow must not produce arbitrageable drift, or the simulator is exploitable.	Returns near-unpredictable despite persistent signs. Lo & MacKinlay (1988) .

Facts 2 and 3: Trajectory and Decay, $N = 6$ Proxy

Quantity	Measured	Reference
Fact 2 — trajectory in executed volume ϕ		
$y(0.25)/y(1)$	0.27	—
$y(0.50)/y(1)$	0.39	0.87 concave ϕ^{δ_2} ; 0.71 $\sqrt{\phi}$; 0.50 linear
$y(0.75)/y(1)$	0.54	—
Fact 2 — dependence on execution time T		
Duration exponent γ	0.247	0.288 from $0.5 - \delta_2$ 0.522 from $H - \delta_2\nu$
Volume-clock exponent ν	0.194	1.0 for a linear clock
Fact 3 — decay after completion, rescaled clock $z = t/T$		
Propagator exponent β	0.142	0.114 reported for DOGE in the notebook
Peak location z	1.04	1.0 if impact peaks at completion
Fit R^2 from the peak	0.986	—

The trajectory is *linear*, not concave. That is a property of the proxy on this instrument, not of the market: the median synthetic parent holds 66% of its volume in a single child, so the volume axis is barely resolved. Reported as measured, without correction.

Fact 11: Diffusivity under Persistent Flow

Fact 11 - diffusivity under persistent flow | Normal (18-24 Jul 2026)

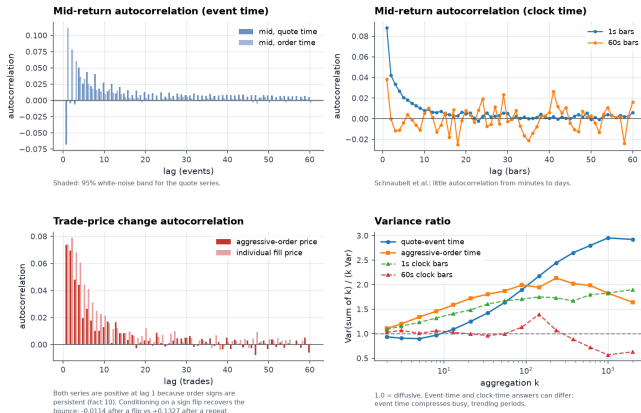


Figure 15: Variance ratio is 0.90 at short quote-event lags (bid-ask bounce) but 2.66 at $k \geq 100$; on 60 s bars it is 0.94. Adaptive liquidity does not fully absorb the persistent flow at minute scales.

P2 — Realism and Second Order (1/2)

#	Fact	Definition	Why prioritised	Extant finding and source
13	Queue-imbalance predictability	$(q_b - q_a)/(q_b + q_a)$ against the next price move.	The skew signal an Avellaneda–Stoikov maker can quote on. Improves maker realism; does not move the cost estimate.	Predicts the one-tick-ahead move. Gould & Bonart (2016) .
14	Volatility clustering	Autocorrelation of absolute returns across horizons.	The feedback channel for <i>cascading</i> liquidations. Second order for a single liquidation, first order if cascades are simulated.	Slowly decaying at minute and hour scales. Schnaubelt et al. (2019) ; Cont (2001).
12	Selective liquidity taking	Book state in the minutes before unusually large aggressive orders.	A forced liquidation cannot time its entry, so this only bounds the gap against voluntary trading.	Large trades are timed into favourable liquidity 2–3 minutes ahead. Schnaubelt et al. (2019) .

Fact 13: Queue-Imbalance Predictability

Fact 13 - queue imbalance | Normal (18-24 Jul 2026)

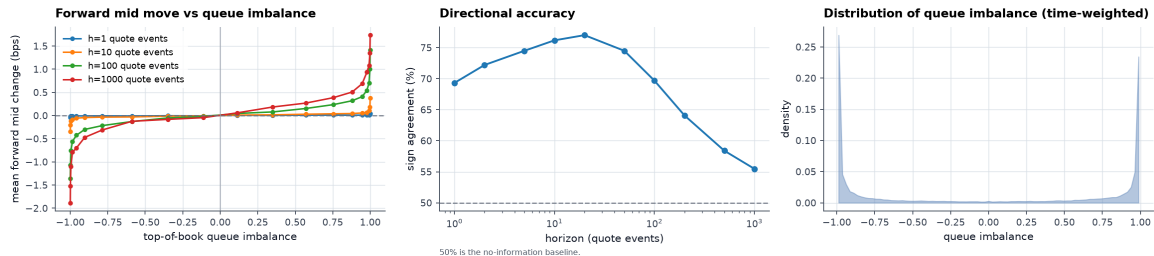


Figure 16: Queue imbalance carries directional information about the next price move; accuracy is reported against the majority-class baseline and with coverage.

Fact 14: Volatility Clustering

Fact 14 - volatility clustering | Normal (18-24 Jul 2026)

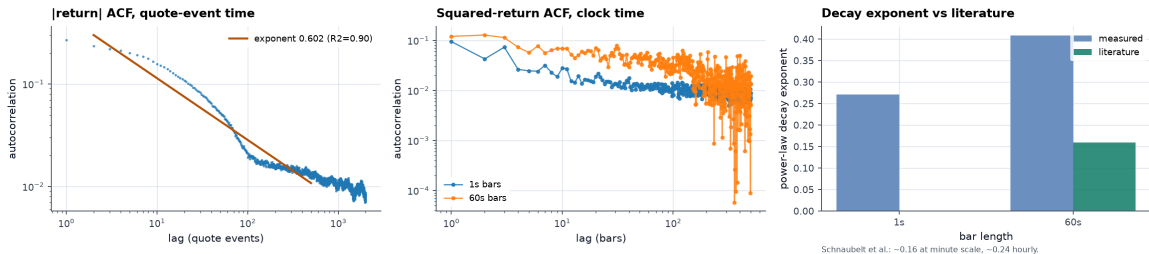


Figure 17: Absolute returns stay autocorrelated far beyond the return autocorrelation itself, at both minute and hour scales.

Fact 12: Selective Liquidity Taking

Fact 12 - selective liquidity taking | Normal (18-24 Jul 2026)

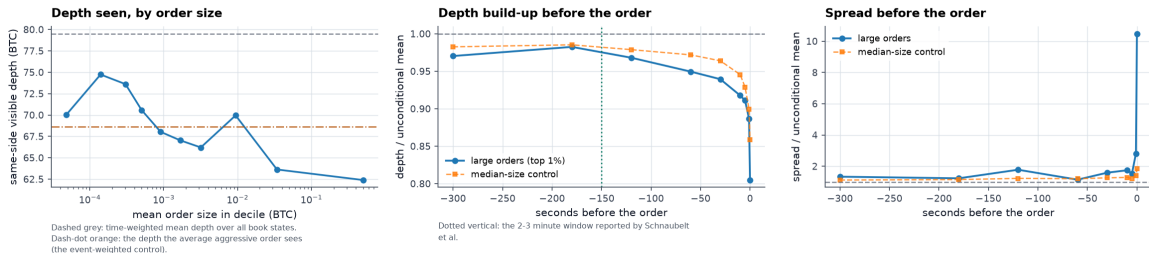


Figure 18: Book state in the minutes before unusually large aggressive orders, against a time-weighted baseline.

P2 — Realism and Second Order (2/2)

#	Fact	Definition	Why prioritised	Extant finding and source
18	Price-diffusion scaling	$\sigma(\tau) = \text{std}(r_\tau) \sim \tau^H$ over 0.2–60 s.	Fixes the volatility scale that impact is quoted in. Affects units, not the mechanism.	$H \approx 0.5$ over the diffusive range, flatter at sub-second lags. Lo & MacKinlay (1988) .
22	Order-timing regularity	Periodicity of arrival times and wall-clock anchoring.	TWAP is itself a clock and the market has one too, so scheduled children can meet a crowd. Refines timing, not size.	Activity concentrates at one-second intervals (Muravyev & Picard, 2022); spot BTC surges in the first second of the minute (Shynkevich, 2026).
17	24/7 intraday structure	Hourly profile of volume, spread and depth.	Face validity only: a run lasting minutes to hours is unaffected by hour-of-day structure.	No equity-style U-shape, but hour-of-day variation persists. Schnaubelt et al. (2019) .

Fact 18: Price-Diffusion Scaling

Fact 18 - price-diffusion scaling | Normal (18-24 Jul 2026)

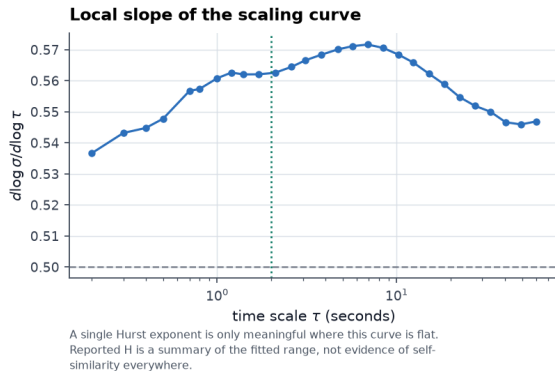
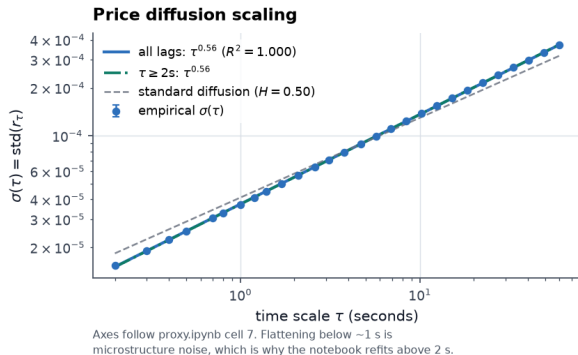


Figure 19: $\sigma(\tau) \sim \tau^H$ with $H = 0.563$ for $\tau \geq 2$ s, slightly super-diffusive; the curve flattens below 1 s where microstructure noise dominates.

Fact 22: Order-Timing Regularity

Fact 22 - order-timing regularity | Normal (18-24 Jul 2026)

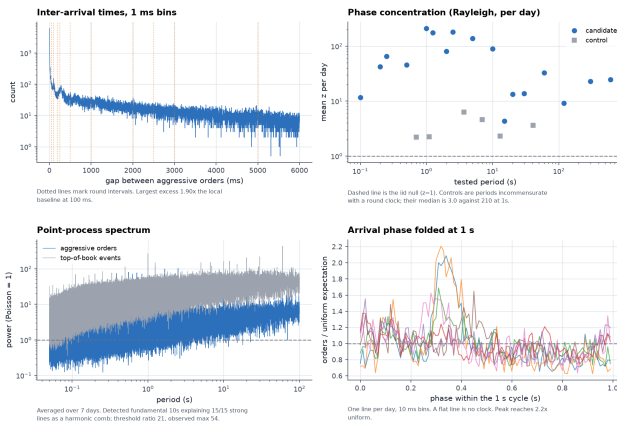


Figure 20: A real execution clock: per-day Rayleigh $z = 210$ at 1s against incommensurate controls at 2–6. Every strong spectral line is an exact multiple of 0.1 Hz, a harmonic comb on a 10s fundamental.

Fact 22: Clock Periodicity in the Published Conventions

Fact 22b - clock periodicity, published conventions | Normal (18-24 Jul 2026)

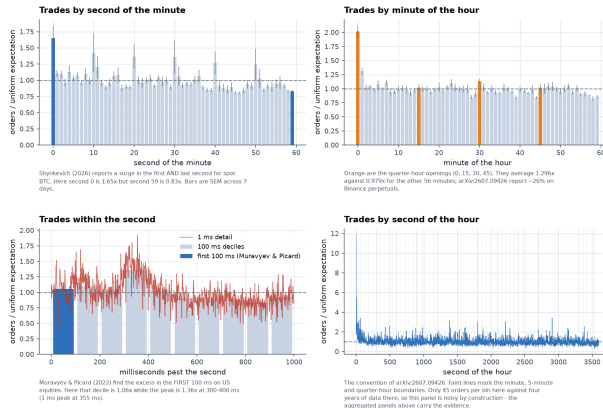


Figure 21: Second 0 of the minute is $1.65\times$ and minute 0 of the hour $2.02\times$ uniform. The within-second peak sits at 300–400 ms, not in the first 100 ms as reported for US equities — consistent with transit latency to a non-located venue.

Fact 17: 24/7 Time-of-Day Structure

Fact 17 - 24/7 intraday structure | Normal (18-24 Jul 2026)



Figure 22: No equity-style U-shape (edge-to-middle volume ratio 0.62), but hourly volume still varies by a factor of 5.6 and spread by 2.4.

Summary: Normal Week

Matches the literature	Does not
Sign long memory, $\gamma = 0.54$	Metaorder concavity far below square root: $\delta \approx 0.21$ – 0.23 against 0.5
$\mathcal{R}(\ell)$ rising and settling	Execution trajectory linear, not concave (0.39 against 0.71)
Cost rising with size; symmetric book	OFI dominates trade-flow imbalance (0.52 against 0.21 at 10 s)
Volatility clustering; heavy size tail	$\mathcal{R}^+(1) > \mathcal{R}^-(1)$ reverses once the spread faced is controlled
Queue imbalance predictive	No U-shape, but hour-of-day variation is large
Over-dispersed arrivals	Within-second peak at 300 – 400 ms, not the first 100 ms

Caveat on Facts 1–3. The proxy assembles parents at random, so a group of six trades is typically one large trade plus small ones: the median parent holds 66% of its volume in a single child. The volume-progress axis is therefore barely resolved, and the linear trajectory should be read as "not measurable on this instrument", not as a market property.

Implications for the Simulator

Design choice	What this data says
Taker arrival process	Do not use Poisson. Counts are over-dispersed by $3\times$ at 1 s and $28\times$ at 300 s, and interarrivals have a heavy tail. A self-exciting (Hawkes) or burst-and-gap process is the minimum that reproduces this.
Taker order sizes	Heavy-tailed: p99/median $\approx 700\times$, and the largest 1% of orders carry 40% of volume. A single characteristic size will understate queue depletion.
Arrival timing	Order flow is anchored to the wall clock: second 0 of the minute is $1.65\times$ uniform, minute 0 of the hour $2.02\times$, with a within-second peak at 300–400 ms. A TWAP child order landing on the round second meets a crowd.
Market-maker quoting	The book is one tick wide 97.6% of the time but only 80.2% of updates; wide spreads are short-lived (16 ms mean dwell against 164 ms). Quote width must be modelled in time, not in events.
Impact calibration target	Signed impact is concave but far below square root on this proxy ($\delta \approx 0.21\text{--}0.23$). Do not hard-code $\delta = 0.5$; treat it as a parameter and report the proxy caveat with it.
Price-flow coupling	Book-event imbalance dominates trade-flow imbalance ($R^2 = 0.52$ against 0.21 at 10 s), and the relation peaks near 60 s. A trades-only impact model leaves most of the explainable move on the table.

Appendix

Normal versus Stress Week

Normal: 18–24 July 2026 · Stress: 26 July – 2 August 2026

By 31 July, options positioning had turned bearish and spot sold off into month end.

Reconstruction Validation

Data quality and book-reconstruction validation

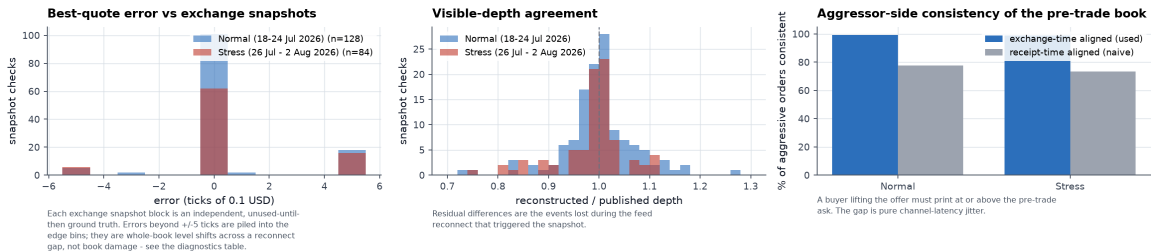


Figure 23: Reconstructed quotes against the exchange's own snapshot blocks: 107 free out-of-sample checks across both weeks, median error 0 ticks.

Facts 1–3: Metaorder Impact, $N = 6$ Proxy

Quantity ($N = 6$, mean of 3 seeds)	Normal	Stress
Parents measured	35,834	43,088
Median children per parent	3	3
Median duration (s)	30.0	27.7
Share with positive signed impact	0.556	0.610
Broken power law δ_2	0.212	0.196
Single-slope δ	0.230	0.216
Prefactor Y	0.061	0.057
Duration exponent γ	0.247	0.201
Volume-clock exponent ν	0.194	0.118
Decay exponent β	0.142	0.142
Hurst H ($\tau \geq 2$ s)	0.563	0.567
Trajectory $y(0.5)/y(1)$	0.389	0.397

Reported as a table because the per-regime comparison figure is generated only for the legacy run-based proxy. Seed spread is construction sensitivity on one market sample, not a confidence interval.

Fact 5: Liquidity Cost and Depth

Fact 5 - liquidity cost and depth by regime

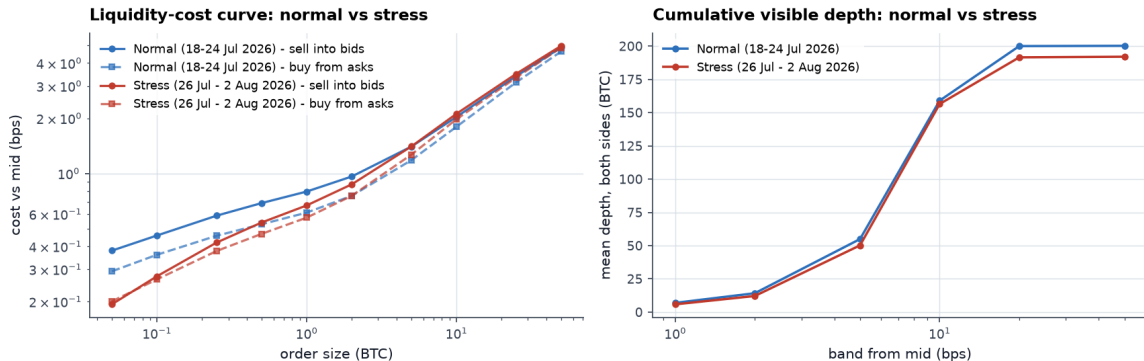


Figure 24: Cost and visible depth, normal against stress.

Fact 6: Spread Distribution

Fact 6 - spread by regime

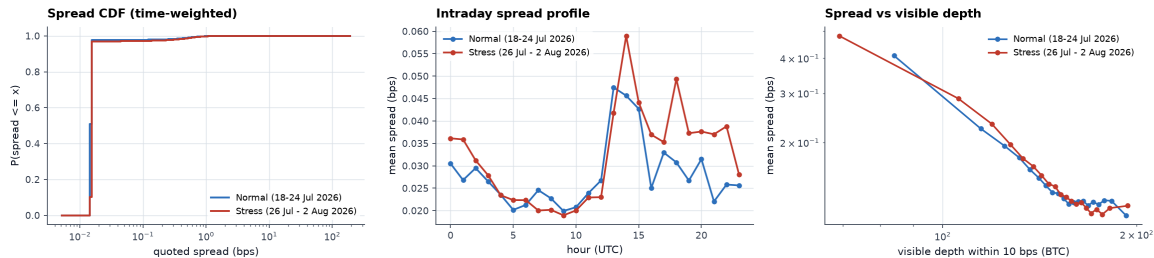
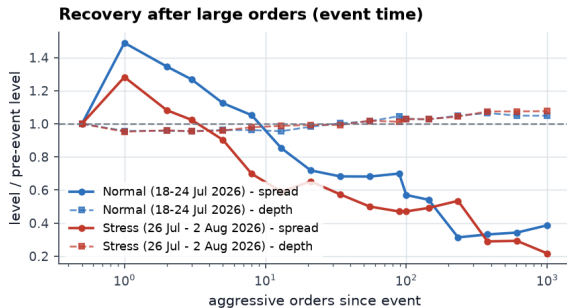


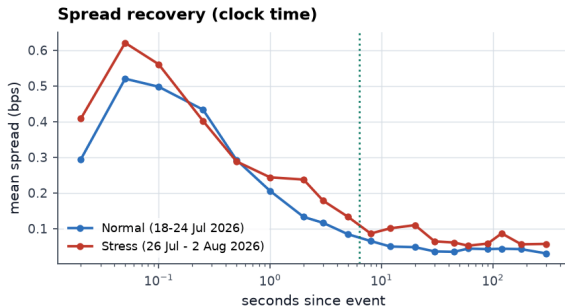
Figure 25: Spread widens in the stress week: mean 2.05 ticks against 1.83.

Fact 7: Liquidity Resiliency

Fact 7 - resiliency by regime



Curves settle below 1 for the spread because large orders arrive when the spread is already wide; the pre-event level is not the steady state. See the per-regime panels for the unconditional mean.



Dotted vertical line: 6.3 s Coinbase BTC/USD literature target.

Figure 26: Recovery after large aggressive orders, both regimes.

Fact 8: Order-Flow Imbalance

Fact 8 - order-flow imbalance by regime

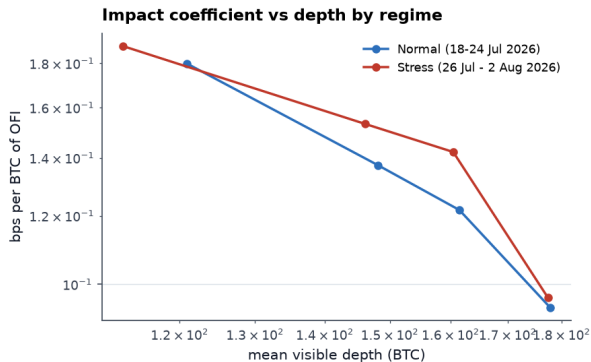
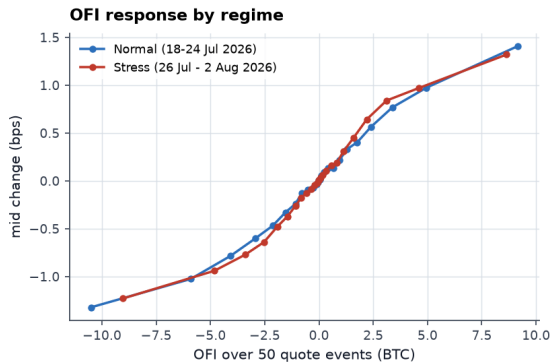


Figure 27: OFI response and its depth dependence.

Fact 8b: OFI versus TFI

Fact 8b - OFI vs TFI by regime

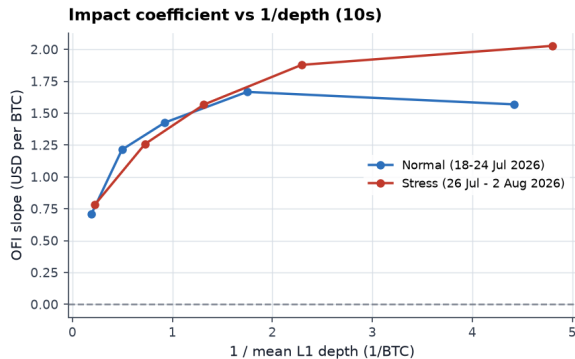
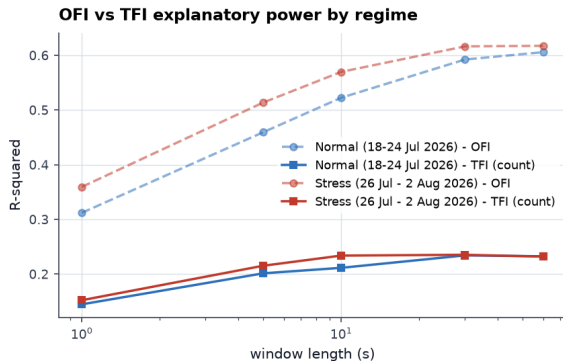


Figure 28: OFI leads trade-flow imbalance on every horizon and in both regimes.

Fact 8c: Real-Time OFI

Fact 8c - real-time OFI by regime

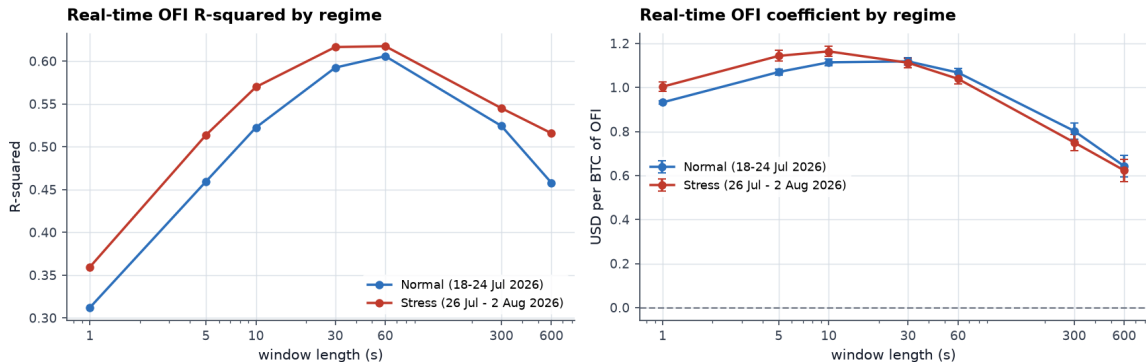


Figure 29: R^2 peaks near 60 s in both weeks (0.61 normal, 0.62 stress) and decays beyond.

Fact 9: Trade Response Function

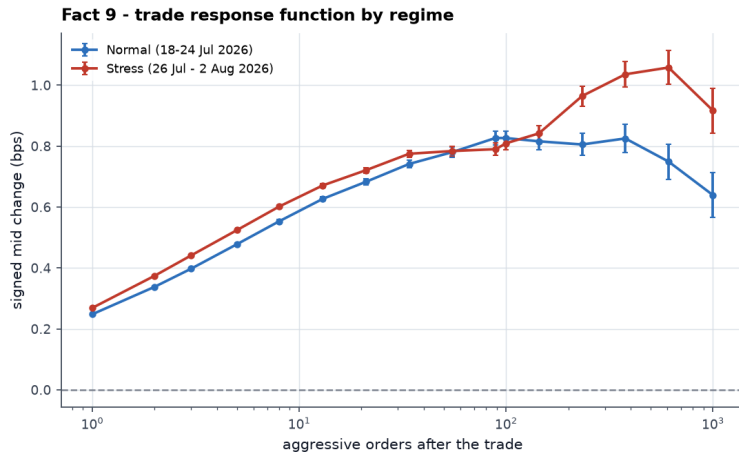


Figure 30: Response function by regime.

Fact 9: Conditional Response

Fact 9b - conditioned trade response by regime

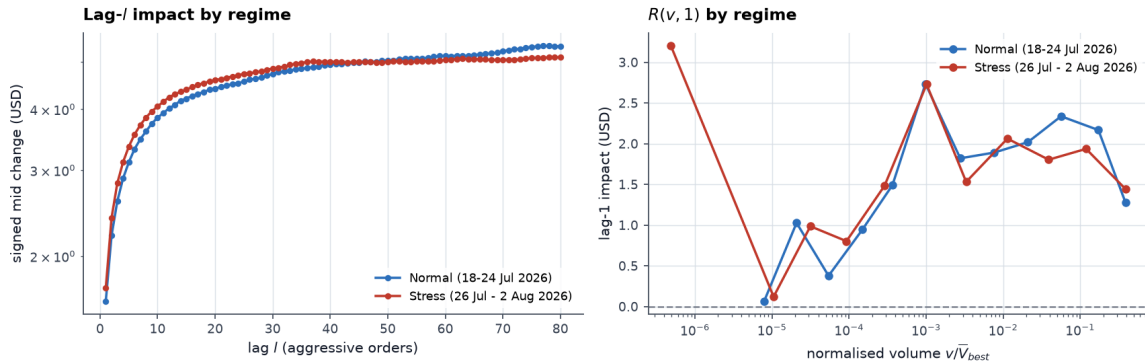


Figure 31: $\mathcal{R}(\ell)$, $\mathcal{R}(v, 1)$ and the sign-conditioned split.

Fact 10: Order-Sign Long Memory

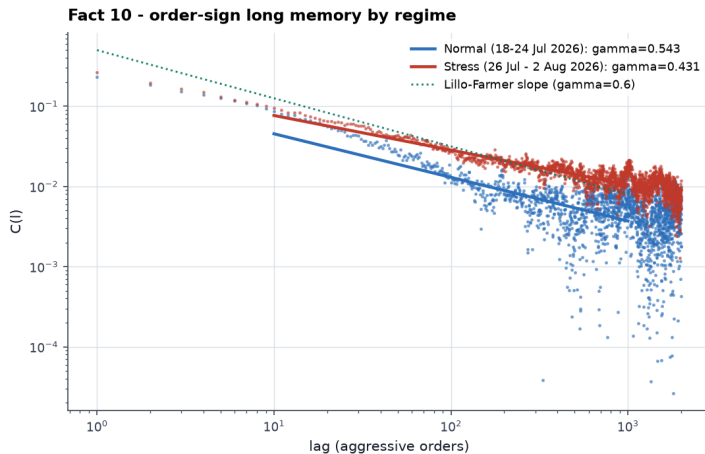


Figure 32: Sign autocorrelation: $\gamma = 0.54$ normal, 0.43 stress.

Fact 11: Diffusivity

Fact 11 - price diffusivity by regime

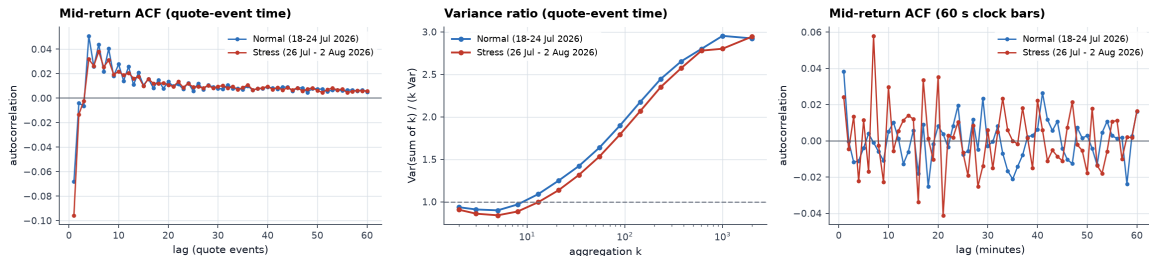


Figure 33: Variance ratios and return autocorrelation by regime.

Fact 12: Selective Liquidity Taking

Fact 12 - selective liquidity taking by regime

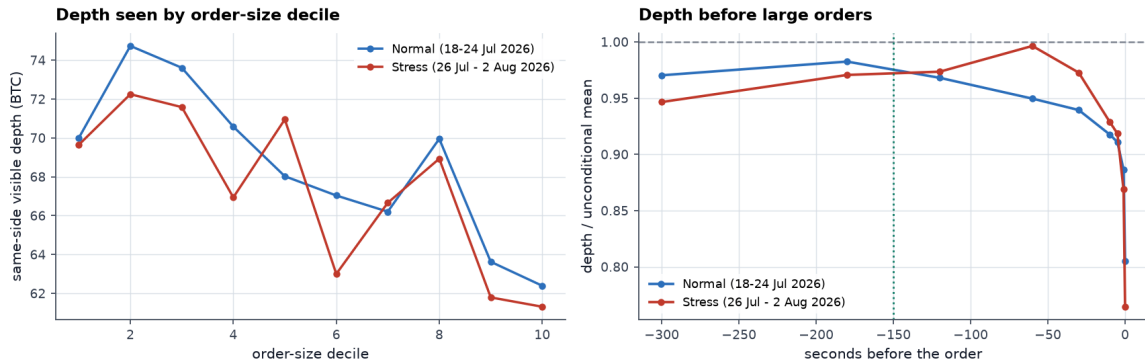


Figure 34: Pre-trade liquidity around large aggressive orders.

Fact 13: Queue Imbalance

Fact 13 - queue imbalance by regime

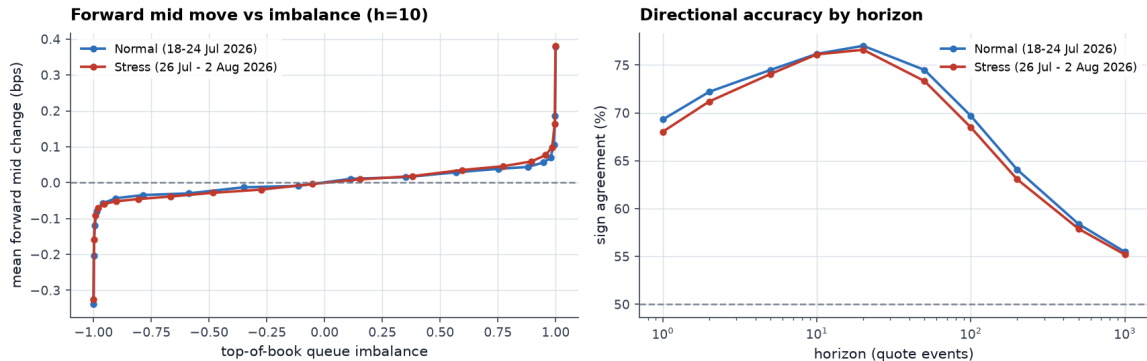


Figure 35: Queue-imbalance predictability by regime.

Fact 14: Volatility Clustering

Fact 14 - volatility clustering by regime

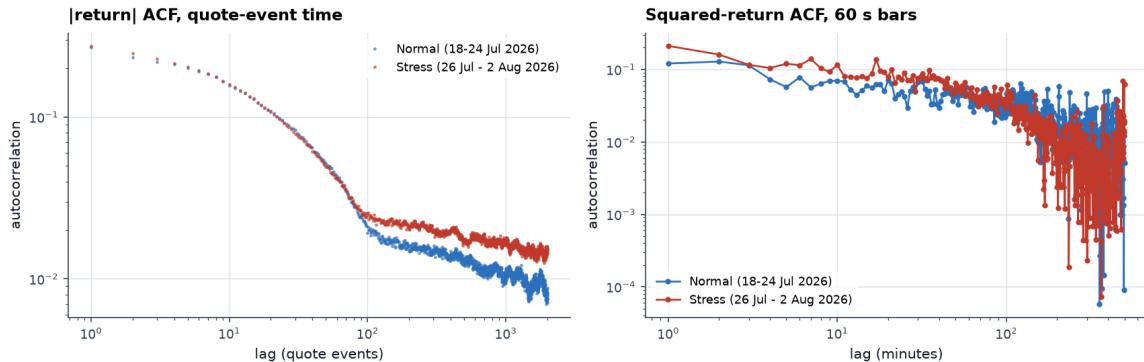


Figure 36: Absolute-return autocorrelation by regime.

Fact 16: Trade-Size Distribution

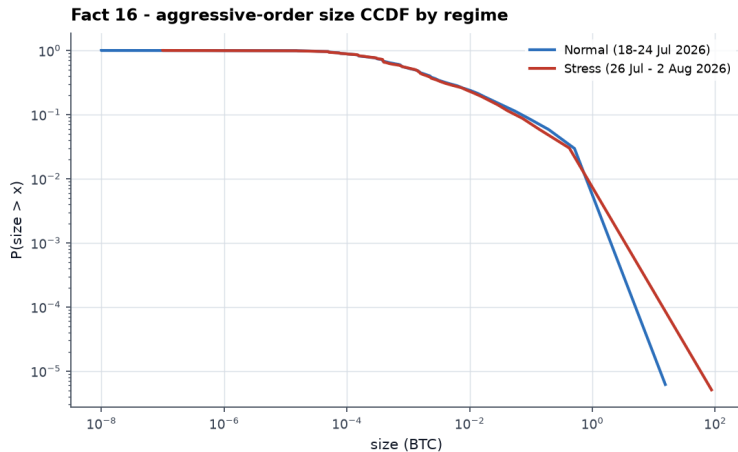


Figure 37: Aggressive-order and fill size distributions.

Fact 17: Time-of-Day Structure

Fact 17 - 24/7 intraday structure by regime

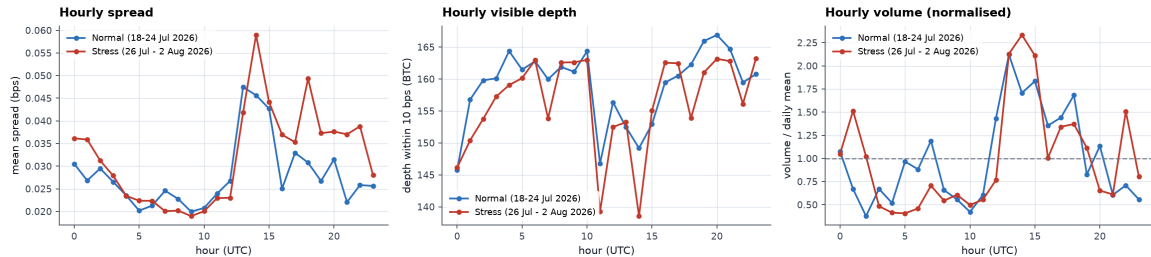


Figure 38: Hourly volume, spread and depth profiles.

Fact 18: Price-Diffusion Scaling

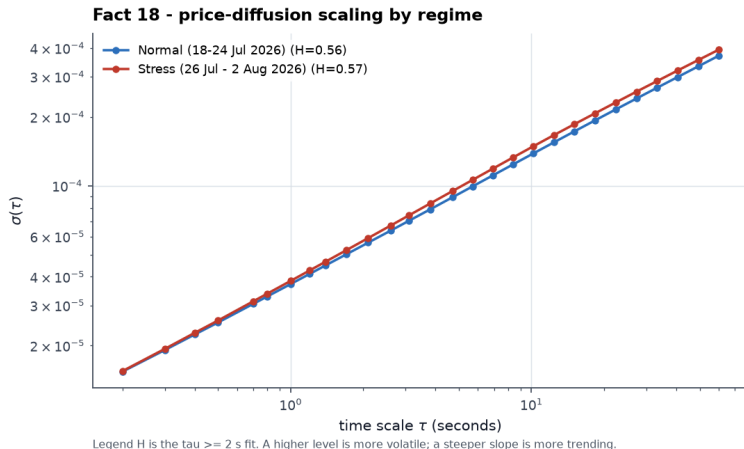


Figure 39: $H = 0.563$ normal against 0.567 stress.

Fact 19: Event Arrivals

Fact 19 - arrivals by regime

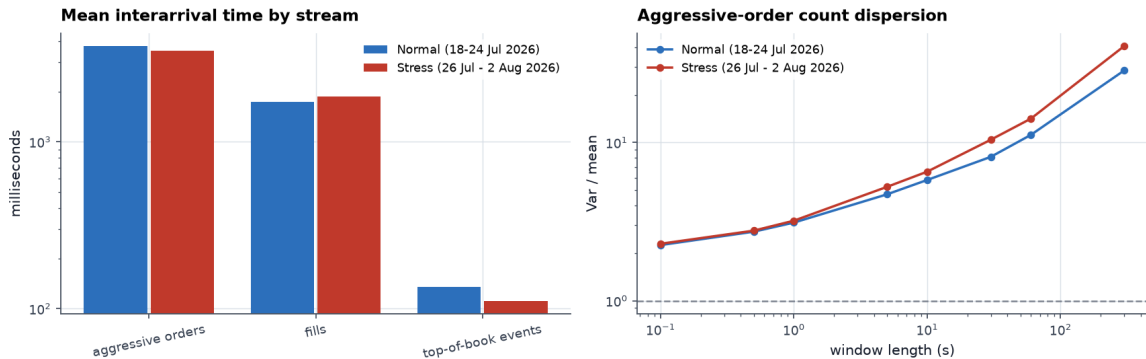


Figure 40: Interarrival distributions and count dispersion.

Fact 20: Book Shape

Fact 20 - book shape by regime

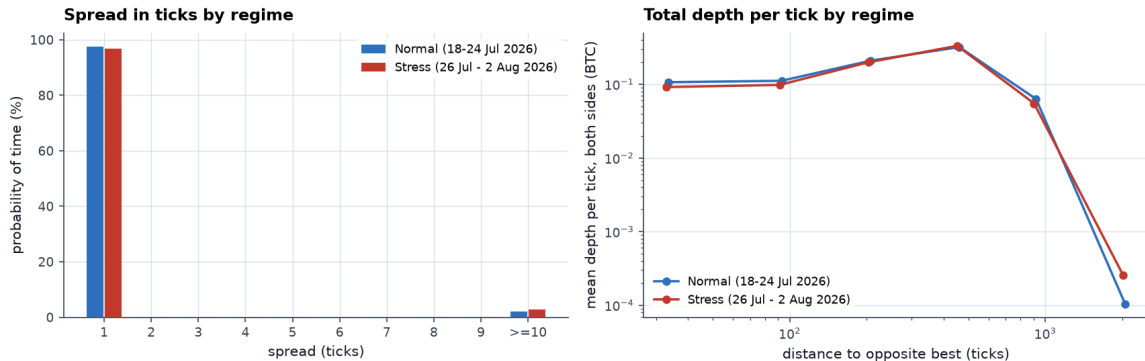


Figure 41: Tick spread and depth-per-tick profile by regime.

Fact 22: Order-Timing Regularity

Fact 22 - order-timing regularity by regime

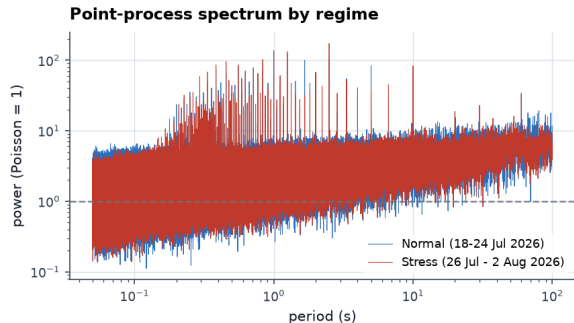
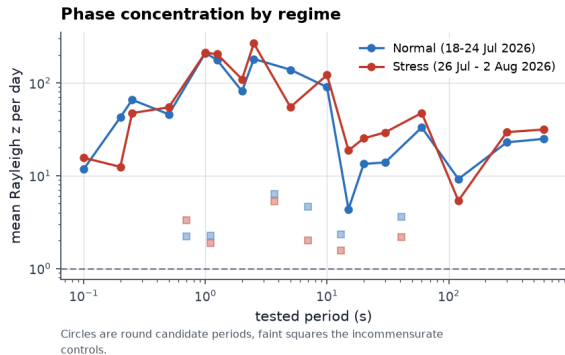


Figure 42: Phase concentration and point-process spectrum, both weeks.

Fact 22: Clock Periodicity

Fact 22b - clock periodicity by regime

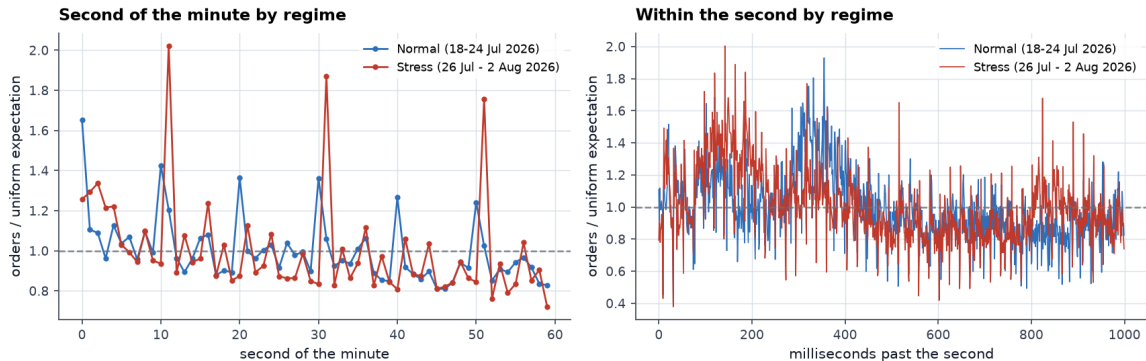


Figure 43: Second-of-minute and within-second profiles by regime.

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